DRAWING DESCRIPTIONS — PATENT C

Method and System for Autonomous Self-Regulation and Cognitive Resynchronization in Neural Processing Systems During Disconnection from External Control Layers

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a system architecture block diagram showing the neural processing system with an external cognitive control layer 10, a heartbeat watchdog 12, a cognitive-driven operation block 14, an autonomous fallback controller 16, a step buffer 18, a state snapshot store 20, a spiking neural network 22, and an energy harvester 24, illustrating the transition mechanism between connected and autonomous operation.

[0012] FIG. 2 is a state transition diagram showing a connected state 26, an autonomous state 28, and a recovering state 30, with a first transition arrow 32 from connected to autonomous on heartbeat timeout, a second transition arrow 34 from autonomous to recovering on resync invocation, a third transition arrow 36 from recovering to connected on resync completion, a fourth transition arrow 38 as a direct reconnection path from autonomous to connected, a self-loop arrow 40 for autonomous steps, and a state legend box 42.

[0013] FIG. 3 is a graph showing the energy-aware modulation curve plotted on a y-axis 44 of modulation value and an x-axis 48 of system energy level, with a behavioral effects table 46, a CRITICAL plateau 50 from 0 to 15 mWh at modulation -0.4, a LOW plateau 52 from 15 to 30 mWh at modulation -0.1, a NORMAL plateau 54 from 30 to 80 mWh at modulation 0.0, and a SURPLUS plateau 56 from 80+ mWh at modulation +0.3.

[0014] FIG. 4 is a signal diagram showing an input signal graph 58 depicting the output waveform of the autonomous input generator including a circadian base signal and attention bursts 60, an autonomous input formula box 62 specifying the signal equation, and an energy recovery graph 64 showing energy level trajectory across modulation zones.

[0015] FIG. 5 is a structural diagram of a resynchronization payload 66 transmitted to the cognitive control layer upon reconnection, comprising a summary statistics section 68, an energy delta detail section 70, a full step buffer section 72, and a decision diamond 74 that determines whether Level 1 summary or Level 2 full buffer is transmitted.

[0016] FIG. 6 is a set of timeline diagrams showing a normal reconnection timeline 76, an extended disconnection timeline 78 with snapshot icons 80 and an energy level indicator 82, a crash recovery timeline 84, a clean shutdown timeline 86, and a recovery guarantees box 88 annotating data integrity across all four recovery scenarios.

[0017] FIG. 7 is an end-to-end signal flow diagram showing the complete system during both connected and autonomous operation, with an external cognitive control layer 10, a step buffer 18, a state snapshot store 20, a spiking neural network 22, an energy harvester feedback path 24, environmental sensors 90, a cognitive modulation path 92, motor actuation 94, an autonomous input generator 96, energy-aware self-modulation 98, and a resync path 100, illustrating how the fallback mechanism seamlessly replaces external cognitive modulation with internal energy-aware self-modulation without interrupting neural processing or energy harvesting.

DETAILED DESCRIPTION OF THE DRAWINGS

Reference Numeral Convention: Reference numerals are assigned sequentially across all figures using even-numbered increments starting at 10. Where the same physical component appears in multiple figures (e.g., the spiking neural network 22 in FIG. 1 and FIG. 7), it retains the same numeral for consistency.

FIG. 1 — System Architecture with Fallback

FIG. 1 is a system architecture block diagram showing the complete system including the fallback mechanism. The diagram is arranged in three vertical sections.

The top section depicts an external cognitive control layer 10 drawn as a dashed ellipse and labeled "External Cognitive Control (Claude)."

The external cognitive control layer 10 provides intelligent cognitive decisions to the system. A downward arrow exits the external cognitive

control layer 10 carrying modulation commands and a heartbeat signal implicit via tool calls. A dashed outline around the external cognitive control layer 10 indicates that it may become unavailable.

The middle section contains a heartbeat watchdog 12 depicted as a rectangle.

The heartbeat watchdog 12 monitors the time elapsed since the last cognitive input, with a timeout threshold of 30 seconds and a check interval of 5 seconds. Two output paths exit the heartbeat watchdog 12: a solid left arrow labeled "active" leading to connected mode, and a dashed right arrow labeled "timeout" leading to autonomous mode.

On the left path, a cognitive-driven operation block 14 is depicted as a rectangle. The cognitive-driven operation block 14 receives input controlled by Claude. An arrow from the cognitive-driven operation block 14 proceeds downward to the neural processing section.

On the right path, an autonomous fallback controller 16 is depicted as a rectangle with bold outline. The autonomous fallback controller 16 generates

self-regulated input comprising a circadian base signal plus stochastic attention bursts, and applies energy-aware modulation in the range -0.4 to +0.3. An arrow from the autonomous fallback controller 16 proceeds downward to a step buffer 18 depicted as a rectangle, and then to the neural processing section. Additionally, the autonomous fallback controller 16 outputs to a state snapshot store 20 depicted as a cylinder shape.

The bottom section contains a spiking neural network 22 and an energy harvester 24, depicted together as a large rectangle labeled "Spiking Neural Network + Energy Harvester." The spiking neural network 22 and energy harvester 24 provide continuous operation regardless of whether the control source is the cognitive-driven operation block 14 or the autonomous fallback controller 16. A key annotation states: "Seamless transition: SNN continues processing without interruption. The control source changes; the computation continues."

FIG. 2 — State Transition Diagram

FIG. 2 is a finite state machine diagram with three states and labeled transitions between them.

A connected state 26 is depicted on the left with a bold outline. The connected state 26 is labeled "CONNECTED" with sublabel "Cognitive layer active." A start indicator points to the connected state 26 identifying it as the initial state.

An autonomous state 28 is depicted on the right. The autonomous state 28 is labeled "AUTONOMOUS" with sublabel "Fallback active" and "Self-regulated."

A recovering state 30 is depicted at the bottom center. The recovering state 30 is labeled "RECOVERING" with sublabel "Resync in progress."

A first transition arrow 32 from the connected state 26 to the autonomous state 28 is labeled "Heartbeat timeout (>30s no tool call)."

A second transition arrow 34 from the autonomous state 28 to the

recovering state 30 is labeled "resync() called."

A third transition arrow 36 from the recovering state 30 to the connected

state 26 is labeled "Resync complete."

A fourth transition arrow 38 shown as a dashed line from the autonomous

state 28 directly to the connected state 26 is labeled "Any tool call (direct reconnect)."

A self-loop arrow 40 on the autonomous state 28 is labeled "Each step (max 10K)."

A state legend box 42 at the bottom provides transition annotations explaining the meaning of each state and its conditions.

FIG. 3 — Energy-Aware Modulation Curve

FIG. 3 is a graph showing the step function that maps energy level to autonomous modulation value.

The y-axis 44 is labeled "Modulation Value" with tick marks at -0.4, -0.1, 0.0, and +0.3 corresponding to the actual modulation output values. The x-axis 48 is labeled "System Energy Level (mWh)" ranging from 0 to 100+ with tick marks at 0, 15, 30, 80, and 100 mWh.

The energy modulation curve plotted on axes 44 and 48 comprises four plateaus connected by vertical step transitions, with vertical dashed lines at the energy thresholds of 15, 30, and 80 mWh.

A first plateau 50 spans from 0 to 15 mWh at modulation -0.4, labeled "CRITICAL." This region represents maximum conservation mode where neural activity is inhibited to preserve energy.

A second plateau 52 spans from 15 to 30 mWh at modulation -0.1, labeled "LOW." This region represents cautious operation with slightly reduced activity.

A third plateau 54 spans from 30 to 80 mWh at modulation 0.0, labeled

"NORMAL." This region represents standard autonomous processing with neutral modulation.

A fourth plateau 56 spans from 80+ mWh at modulation +0.3, labeled

"SURPLUS." This region represents opportunistic exploration where excess energy enables increased activity.

A behavioral effects table 46 appears below the graph and lists the

modulation value and behavioral description for each energy region:

CRITICAL (-0.4) maximum conservation, LOW (-0.1) cautious operation, NORMAL (0.0) standard processing, SURPLUS (+0.3) opportunistic exploration.

FIG. 4 — Autonomous Input Generator Output

FIG. 4 is a signal diagram showing the time-series waveform of the self-generated input signal.

An input signal graph 58 occupies the upper portion of the figure with

x-axis "Autonomous Step Number" ranging from 0 to 500 and y-axis "Input Value" ranging from 0.0 to 1.0.

The input signal graph 58 contains superimposed waveform components.

A circadian base signal is shown as a smooth sinusoidal curve. Superimposed on the base signal are attention bursts 60 shown as sharp upward spikes occurring randomly at approximately 10% of steps, each burst 60 adding +0.2 above the base signal.

An autonomous input formula box 62 shows the equation governing the composite signal generation: input = 0.5 + 0.3 x sin(2 x pi x step / period) + burst, with period = 500 steps, burst probability = 10%, and burst amplitude = +0.2.

Below the input signal graph 58, an energy recovery graph 64 shows the corresponding energy level over the same time period. The energy recovery graph 64 has y-axis "Energy (mWh)" ranging from 0 to 100 and shows a monotonic recovery trajectory starting in the CRITICAL zone and rising through all four modulation zones to the capacity ceiling. Three horizontal

dashed lines appear at 15, 30, and 80 mWh thresholds with zone labels: CRITICAL, LOW, NORMAL, SURPLUS.

FIG. 5 — Resynchronization Payload Structure

FIG. 5 is a structural diagram showing the hierarchical data structure of the resync payload.

A resynchronization payload 66 is depicted as a large top-level rectangle containing three main sections.

A summary statistics section 68 is depicted as a rectangle within the resynchronization payload 66. The summary statistics section 68 contains fields: "steps_autonomous: [integer]," "energy_delta: [float] mWh," "min_energy / max_energy / mean_energy," and "total_spikes / mean_spikes_per_step." The summary statistics section 68 is labeled "SUMMARY STATISTICS (always included)" with size annotation "~200 bytes."

An energy delta detail section 70 is depicted as a rectangle within the

resynchronization payload 66. The energy delta detail section 70 contains fields: "energy_start: [float]" and "energy_end: [float]." An arrow from the section leads to annotation: "Positive = gained."

A full step buffer section 72 is depicted as a larger dashed rectangle within the resynchronization payload 66 with a repeating row structure showing individual step records: "Step 0: {input, modulation, spikes, energy, temp, pattern}" through "Step N: {input, modulation, spikes, energy, temp, pattern}." The full step buffer section 72 is labeled "FULL STEP BUFFER (optional)" with size annotation "~100 bytes x N steps" and condition **"included when cognitive layer requests history."**

Two granularity levels are annotated: a bracket labeled "Level 1: Summary" pointing to the summary statistics section 68, and a bracket labeled "Level 2: Full" pointing to the full step buffer section 72.

A decision diamond 74 at the bottom of the figure asks "Full buffer

requested?" A YES arrow leads to "Level 2" and a NO arrow leads to "Level 1."

FIG. 6 — Recovery Timeline Diagrams

FIG. 6 shows four horizontal timeline diagrams stacked vertically, each illustrating a different recovery scenario.

A normal reconnection timeline 76 depicts the first scenario. The normal reconnection timeline 76 progresses from left to right through five phases: Connected, Timeout (30s), Autonomous (N steps), Resync, and Connected again.

Annotations include: "Claude silent" at the transition to timeout, "Fallback engages" at the transition to autonomous mode, and "resync()" at the transition to resync.

An extended disconnection timeline 78 depicts the second scenario. The extended disconnection timeline 78 progresses through: Connected, Timeout, Autonomous (many steps), Resync, and Connected. Snapshot icons 80 appear as

black squares along the autonomous phase indicating periodic state serialization every 50 steps. An energy level indicator 82 appears as a bar gauge showing the energy state during extended autonomous operation.

A crash recovery timeline 84 depicts the third scenario. The crash recovery timeline 84 progresses through: Connected, CRASH (bold X marker), Restart (process relaunch), Snapshot Load (reads latest.json), and Connected (state **restored). Annotation notes "Max 50 steps lost."**

A clean shutdown timeline 86 depicts the fourth scenario. The clean shutdown timeline 86 progresses through: Connected, Autonomous, EOF (stdin closed), and Shutdown (final snapshot). Annotation notes "State preserved for next session."

A recovery guarantees box 88 provides common annotations for all four timelines: solid line segments indicate "System actively processing," dashed line segments indicate "System in transition," and bold markers at each state

change indicate "No data loss at state change."

FIG. 7 — End-to-End Signal Flow: Connected vs. Autonomous Operation

FIG. 7 is an end-to-end signal flow diagram showing the complete system in a split-view format, with connected operation on top and autonomous operation on bottom, sharing the same core processing pipeline.

In the top half labeled "CONNECTED MODE," a cognitive layer 10 is depicted as a dashed ellipse at top. A cognitive modulation path 92 carries commands downward to the processing pipeline. The connected-mode processing pipeline flows left to right: sensors 90 receive environmental input, feed into the spiking neural network core 22, which feeds into motor actuation 94. An energy harvester feedback path 24 returns from the motor actuation 94 back to the spiking neural network 22 as dashed arrows.

A horizontal dashed mode boundary divides the connected and autonomous sections, labeled "MODE BOUNDARY."

In the bottom half labeled "AUTONOMOUS MODE (Fallback)," an autonomous input

generator 96 replaces the sensors 90, containing "Circadian + Burst" input.

The autonomous-mode processing pipeline flows: autonomous input generator 96

feeds into the spiking neural network core 22, which is modulated by

energy-aware self-modulation 98 (labeled "ENERGY MODULATION (4-zone)").

Additional components in autonomous mode include a state buffer 18 depicted

as an ellipse, and a state snapshot store 20 depicted as an ellipse, both

receiving step data from the spiking neural network 22.

On the right side of the figure, a resync path 100 shows a dashed arrow from

the autonomous section back to the cognitive layer 10, labeled "resync()."

A mode selection logic box at the bottom explains: in connected mode, input

comes from sensors 90 and modulation from Claude via the cognitive modulation

path 92; in autonomous mode, input comes from the autonomous input

generator 96 and modulation from the energy-aware self-modulation 98. The

buffer 18 and snapshot store 20 are active during autonomous mode. The feedback path 100 transmits the resync payload on reconnection.

GENERAL DRAWING REQUIREMENTS (per 37 CFR 1.84)

- Paper size: 8.5 x 11 inches (US Letter)

- Top margin: 1 inch

- Left margin: 1 inch

- Right margin: 5/8 inch

- Bottom margin: 3/8 inch

- Black lines on white background ONLY

- No color, no grayscale shading (use hatching patterns instead)

- Line weight: sufficiently heavy for adequate reproduction
- Each figure labeled: "FIG. 1", "FIG. 2", etc. - Sheet numbers at top center: "1/7", "2/7", etc.

- No frames or borders around drawing area

- All text in figures must be at least 1/8 inch (3.2 mm) high

- Reference numerals must be used consistently across all figures